ACM/IDS 104 APPLIED LINEAR ALGEBRA
PROBLEM SET 5

Please submit your solution as a single PDF file, that contains both the written-up and published code parts, via Gradescope by **9pm Tuesday, November 18**. An example of the submission process is shown here: `https://www.gradescope.com/get_started#student-submission`

- For theoretical problems, please use a pen, not a pencil: it is hard to read scanned submission written by a pencil.
- For coding problems, please convert your MATLAB livescripts (`.mlx`) to PDF by selecting `Live Editor` → `Save` → `Export to PDF` and merge them with the rest of your solution.
- After uploading your submission to Gradescope, please label all pages.

Problem 1.(10 POINTS) APPLICATION OF PROJECTIONS TO APPROXIMATION
Complete Problem 1 in `PS5.mlx`.

Problem 2. (10 POINTS) HERMITE POLYNOMIALS
Consider the following inner product, sometimes called the Hermite inner product[1]:

$$\langle f, g\rangle = \int_{-\infty}^{\infty} f(x)g(x)e^{-\frac{x^2}{2}}\,dx. \tag{1}$$

This inner product appears in probability theory, functional analysis, numerical PDEs, and in physics[2].

Let $f(x) = e^x$, and suppose that we want to approximate this function by a polynomial $p(x) \in \mathcal{P}^{(n)}(\mathbb{R})$ of degree at most n on the real line. Notice that we cannot do this with respect to the standard L^2 inner product (convince yourself that this is the case)[3]. We can do this, however, with respect to the Hermite inner product. In what follows, you can use Wolfram Alpha (or other tools) to compute the arising indefinite integrals.

(a) (3 points) Starting from the standard basis $1, x, x^2$ of the vector space $\mathcal{P}^{(2)}(\mathbb{R})$, use the Gram-Schmidt process to construct an orthogonal basis $h_0(x), h_1(x), h_2(x)$ of $\mathcal{P}^{(2)}(\mathbb{R})$ with respect to the Hermite inner product. The polynomials in this basis are called the *Hermite polynomials*[4].

(b) (3 points) Based on $h_0(x), h_1(x), h_2(x)$ construct an orthonormal basis $H_0(x), H_1(x), H_2(x)$ of $\mathcal{P}^{(2)}(\mathbb{R})$.

(c) (4 points) Find the best approximation $p(x) \in \mathcal{P}^{(2)}(\mathbb{R})$ to $f(x) = e^x$ in the Hermite sense, that is, with respect to the Hermite norm, induced by the Hermite inner product.

Problem 3. (10 POINTS) ORTHOGONAL COMPLEMENTS
In this problem, we will practice finding orthogonal complements in different inner product spaces.

(a) (4 points) Let $W \subset \mathbb{R}^3$ be a subspace spanned by $v_1 = (1, 2, 3)^T$ and $v_2 = (2, 0, 1)^T$. Find the orthogonal complement

(a1) (2 points) $W_1^\perp$ to W with respect to the dot product $\langle x, y\rangle_1 = x_1y_1 + x_2y_2 + x_3y_3$ and

(a2) (2 points) $W_2^\perp$ to W with respect to the weighted product $\langle x, y\rangle_2 = x_1y_1 + 2x_2y_2 + 3x_3y_3$.

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[1]Technically, this inner product is defined on the Hilbert space $L^2(\mathbb{R}, \mu)$, where the measure $d\mu(x) = e^{-\frac{x^2}{2}}\,dx$, which consists of all (measurable) functions $f : \mathbb{R} \to \mathbb{R}$ such that $\int_{-\infty}^{\infty} f(x)^2 e^{-\frac{x^2}{2}}\,dx < \infty$. But for the purpose of this problem, this technicality is not needed. We consider only those functions f and g for which the integral in (1) exists.

[2]In physics, the weight e^{-x^2} is typically used instead of $e^{-\frac{x^2}{2}}$.

[3]We can do this only on the finite interval $[-a, a]$, see Practice Problem 10B.

[4]Hermite polynomials were originally defined by Laplace and studied in detail by Chebyshov. This is yet another manifestation of the Arnold principle: if a notion bears a personal name, then this name is not the name of the discoverer, `https://dsweb.siam.org/The-Magazine/All-Issues/vi-arnold-on-teaching-mathematics`.

(b) (3 points) Let $U = \text{span}(I_n) \subset \mathbb{M}_{n\times n}$ be a subspace spanned by the identity matrix. Find the orthogonal complement $U^\perp$ to U with respect to the Frobenius inner product (introduced in Practice Problem 4B).

(c) (3 points) Let $V = \text{span}\{x, x^3\} \subset \mathcal{P}^{(3)}[-1,1]$ be a subspace of the inner product space $\mathcal{P}^{(3)}[-1,1]$ of polynomials on $[-1,1]$ of degree at most 3 equipped with the L^2 inner product. Find the orthogonal complement $V^\perp$.

Problem 4. (10 points) Real and Complex Eigenbases
Consider the following matrix:

$$A = \begin{bmatrix} 0 & 0 & -1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} \tag{2}$$

(a) (2 points) Is this matrix complete?

(b) (2 points) Does A admit an eigenbasis of $\mathbb{R}^3$? If yes, find it.

(c) (3 points) Does A admit an eigenbasis of $\mathbb{C}^3$? If yes, find it.

(d) (3 points) Construct a basis of $\mathbb{R}^3$ consisting of real and imaginary parts of eigenvectors of A. Is it an eigenbasis?

Problem 5. (10 points) Gershgorin Theorem
Consider the following matrix:

$$B = \begin{bmatrix} 0 & 1 & 0 \\ 0 & 1 & 1 \\ 0 & -1 & 1 \end{bmatrix} \tag{3}$$

(a) (2 points) Find all Gershgorin disks and sketch the Gershgorin domain in the complex plane.

(b) (2 points) Show that the eigenvalues of any matrix A must lie in its *refined* Gershgorin domain, which is defined as the intersection of the Gershgorin domains of A and A^T, $D_A^* = D_A \cap D_{A^T}$.

(c) (2 points) Find and sketch the refined Gershgorin domain D_B^* of matrix B in (3).

(d) (2 points) Find the eigenvalues of B and check that they indeed belong to D_B^*.

(e) (2 points) Give an example of a nonsingular matrix A whose Gershgorin domain contains zero.